



TALENT MANAGEMENT

How Technology Can be Used in Talent Management



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INTRODUCTION

In today's competitive global environment, organisations increasingly rely on technology to enhance talent management (TM). Modern talent management integrates information systems, data analytics, and artificial intelligence to attract, develop, and retain employees in alignment with strategic goals. Talent management technology ranges from digital recruitment platforms and e-HRM systems to advanced analytics and machine learning tools that support decision-making. For example, scholars note that AI is increasingly used in Human Resource Management, particularly within recruitment, and candidates often value AI for innovation, though they express caution when personal data are involved (Tursunbaveva et al. 2025). The core question is: How can technology be used in talent management? This study will examine recent research on this topic by organising insights in key technology areas. It covers digital transformation's impact on TM, e-recruitment and AI-driven hiring, talent analytics and data-driven HR, and emerging technologies such as blockchain. Furthermore, the review will draw on the articles provided and several contemporary studies to provide a comprehensive literature-based answer to the question.

BACKGROUND AND PROBLEM STATEMENT

Organisations face increasing pressure to attract, retain, and develop talent in ways that align with rapid technological change. Whilst effective in the past, traditional approaches to talent management often lack the agility needed to respond to digitalisation, globalisation, and the evolving expectations of employees. This has positioned technology not as a supplementary tool but as a strategic enabler of talent management (Guerra, Danvila-del-Valle, & Méndez-Suaárez, 2023).

One of the most prominent drivers of this transformation is artificial intelligence (AI). According to França (2023), AI-powered tools significantly improve recruitment by enhancing candidate screening and predicting job-fit alignment. In addition, predictive analytics can anticipate employee turnover, enabling organisations to design proactive retention strategies (Căvescu & Popescu, 2025). These technologies highlight how

digital solutions are reshaping not only recruitment but also long-term retention strategies.

Beyond recruitment and retention, digital human resource platforms play a pivotal role in streamlining HR operations and enhancing the employee experience. Integrated HR information systems (HRIS) allow organisations to centralise data, automate routine HR functions, and provide managers with real-time insights into workforce dynamics. Studies have shown that digital HR practices positively influence employee engagement and productivity, which in turn drive organisational performance. This demonstrates that digital HR platforms serve as both an operational necessity and a strategic advantage.

Moreover, the broader shift towards digital transformation has accelerated the adoption of technology in HR. Digitalisation is not limited to HR functions; it influences entire organisational cultures, requiring new skill sets, flexible structures, and innovative leadership approaches (Guerra, Danvila-del-Valle, & Méndez-Suaárez, 2023). Talent management is therefore deeply intertwined with organisational transformation, as the success of digital initiatives often depends on the availability and development of digitally skilled employees.

Yet, these technologies bring challenges. Gonzalez (2019) cautioned that AI and Machine Learning (ML) systems often raise ethical issues, including algorithmic bias, fairness, and transparency. Similarly, Munde (2023) reported that while firms frequently adopt digital tools for payroll and e-learning, many still lack comprehensive solutions for succession planning and strategic talent development. Asfahani (2024) further emphasised that moving from intuition-based to algorithmic HRM requires a balance between efficiency and ethical responsibility.

The literature makes clear that technology is no longer peripheral but central to TM. Whether through AI in recruitment, predictive analytics for retention, HRIS platforms, or blockchain for credential security; technology is reshaping how firms build sustainable talent pipelines.

PROBLEM STATEMENT

The core problem is not whether technology should be used in talent management—it already is—but which types of technologies should be strategically prioritised to ensure global competitiveness while still upholding ethical, human-centred practices.

LITERATURE REVIEW

Digital Transformation and Talent Management

Digital transformation (DT) is fundamentally reshaping how organisations operate and manage people. Technology-driven organisational change involves substantial shifts in processes and culture, which directly affects talent management. Montero Guerra (2023) found that companies undergoing digital transformation see corresponding shifts in how they attract and retain talent. In a study of 314 Spanish firms, they showed that organisational changes brought about by digital transformation are thus seen to influence talent management and to attract and retain talent. In other words, as companies adopt new digital tools and processes, their talent strategies must adapt. For instance, digital maturity often necessitates new skill sets, so HR must use technology to identify and develop those skills.

In the digital context, talent management becomes more integrated with business strategy. For example, Mahroof and Rafi (2024) surveyed organisations in Pakistan and found that effective talent management significantly drives digital transformation, suggesting a virtuous cycle: technology enables better TM, and robust TM in turn accelerates digital change. Their analysis indicates that talent management has a significant positive impact on digital transformation. This implies that investing in talent processes (powered by technology) supports broader transformation goals. Similarly, Sivathanu and Pillai (2019) showed that using technology in TM can improve organisational performance. They reviewed literature and surveyed HR managers in multinational firms, concluding that organisations can achieve better performance by leveraging TM technologies. Their findings emphasize the role of talent metrics and analytics: “organisational performance can be achieved by using technology for talent management,” highlighting analytics and HR business partnerships as key factors.

In summary, digital transformation and talent management are tightly linked. Companies at the forefront of DT adopt integrated talent platforms and data-driven HR, which helps them better align workforce capabilities with evolving needs. The research shows that technology-driven changes in the organisation (e.g., new digital business models) require talent strategies supported by technology (e.g., analytics and digital learning). In turn, robust talent management (including digital skill development and data-driven HR) promotes successful digital transformation.

Recruitment and Selection Technologies

One of the most prominent areas of technology use in TM is recruitment and selection. Online systems, social media, e-recruitment platforms, and AI tools have transformed how organisations find and hire talent. Electronic HRM (e-HRM) systems digitize job postings, applications, and assessments, while artificial intelligence (AI) tools can screen resumes, score candidates, and even conduct initial interviews via chatbots.

In the hospitality and tourism industry, Johnson (2020) illustrated these trends. They discussed how “electronic human resource management (eHRM) and artificial intelligence (AI)” can help firms recruit and select better employees. Their study concluded that eHRM and AI together have the potential to transform how the hospitality and tourism industry recruit and select employees. Specifically, by using online recruitment platforms and AI-driven tools for e-recruiting and e-selection, companies can more efficiently identify qualified candidates, reduce time-to-hire, and even improve retention by matching people to jobs more effectively. However, they caution that organisations must manage the implementation carefully to ensure employee trust and positive outcomes.

Similarly, current research emphasizes the rapidly growing adoption of AI in recruiting. Tursunbayeva (2025) reported that AI is increasingly used in Human Resource Management (HRM), particularly within recruitment, reflecting a broad industry trend. In their study, job candidates generally saw AI positively, appreciating its role in innovation and development, though some expressed concerns about personal data use. This indicates that recruitment technologies must balance efficiency with transparency. In practice, organisations are deploying AI-driven resume screening,

candidate matching algorithms, and automated interview scheduling. These technologies expand sourcing channels (including social media and talent communities) and use data analytics to highlight the best-fit candidates. For example, modern applicant tracking systems can parse applications at scale and recommend candidates based on historical hiring data.

In sum, digital recruitment platforms and AI tools make hiring faster and more scalable. They enable global sourcing, allow for predictive screening, and can personalize candidate engagement. But scholars note that technology should augment—not replace—the human element in recruiting. As Johnson (2020) warned that care is needed to ensure that AI-driven decisions are well received by employees and lead to better employee and organisational outcomes. In practice, this means using AI ethically (avoiding bias) and combining it with human judgment. Overall, technology's use in recruitment—from e-HRM systems to AI analytics—is reshaping talent acquisition by improving reach, speed, and data-driven matching of candidates to roles.

Analytics and Artificial Intelligence in Talent Management

Beyond recruitment, advanced analytics and AI increasingly support broader talent management processes. “Talent analytics” refers to using statistical and data-driven tools to understand and predict workforce behaviors. Organisations collect data on hiring, performance, engagement, retention, and more, then apply analytics to guide decisions. Mahato and Prasad (2024) provided a recent systematic review of talent analytics. They explained that modern talent analytics employs a variety of BI tools and systems to create a more holistic and data-driven view of an organisation's staff. In practice, talent analytics might predict which employees are likely to leave (attrition models), identify skills gaps (through competency assessments), or evaluate training outcomes (by linking learning metrics to performance). By analysing “people data,” HR can make evidence-based decisions on hiring, retaining, or improving workforce results. For example, an organisation may use predictive analytics to flag high performers who should be fast-tracked for development, or to identify underperforming areas needing additional training.

These data-driven approaches have tangible business value. Mahato and Prasad noted that talent analytics tools focus on understanding people data that can inform decisions related to hiring, retaining, or improving results. Analytics is often cited as a strategic enabler of TM. Along with data, artificial intelligence and machine learning (ML) methods are being incorporated into talent analytics. By applying ML to HR data, organisations can uncover complex patterns and make predictions at scale. For instance, Gonzalez (2019) discussed how AI/ML tools are widely adopted for talent assessment. These technologies are used by organisations seeking to identify and hire high-quality job applicants, and can potentially improve selection accuracy. Techniques such as natural language processing allow AI to screen interview transcripts or analyse employee sentiment; ML algorithms can rank candidates by fit; and AI-driven chatbots can manage routine HR inquiries 24/7.

At the same time, Gonzalez (2019) cautioned that empirical evidence on AI/ML in HRM is still limited. Whilst analytics and AI promise better insights, organisations must carefully validate and monitor these systems. Issues such as bias in hiring algorithms and explainability of AI decisions are active research concerns. Interdisciplinary collaboration was suggested to be done amongst psychologists, data scientists, and legal experts to ensure that AI/ML applications are ethical and effective. Practitioners often implement audit processes and human oversight to mitigate these risks.

In operational terms, talent analytics and AI support many TM functions. For performance management, automated analytics dashboards can track key metrics and identify trends (e.g., predicting which teams might fall short of goals). For learning and development, platforms adapt training content based on employee learning patterns. Even simple AI, such as recommendation engines, can tailor career development suggestions. Overall, technology turns raw HR data into actionable knowledge, enabling proactive talent decisions. As Sivathanu and Pillai (2019) emphasized, when organisations harness such technology in TM, better performance outcomes often result. They found that using technology for TM leads to organisational performance gains by enabling talent metrics, analytics, and HR partnerships focused on business results. Thus, analytics and AI make TM more strategic: they move HR from intuition-based decisions to predictive, outcome-driven strategies.

Emerging Technologies in Talent Management

Beyond AI and analytics, a range of emerging technologies is beginning to influence TM. Two notable areas are blockchain and generative AI. These represent newer frontiers of technology that can further improve talent processes.

Blockchain and Security.

Blockchain is a distributed ledger technology that ensures data integrity and transparency. In talent management, blockchain can secure employee records and recruitment credentials. Idris and Bahasoan (2025) reviewed blockchain in TM, focusing on recruitment. They argued that blockchain improves transparency and security in talent management, especially in recruitment and human resource management.

For example, blockchain can store and verify candidate certifications (degrees, licenses), eliminating fraudulent resumes. It can provide a tamper-proof history of employment or training records, allowing recruiters to trust applicant data. Idris and Bahasoan's systematic review found that blockchain solutions can verify candidate certification, simplify the selection process, and maintain the integrity of employee data. Organisations implementing blockchain in HR can thus speed up background checks and reduce operational costs by automating verification. They also noted potential challenges, such as high implementation costs and regulatory issues, but overall saw blockchain as a promising standard for secure, efficient talent acquisition.

Generative AI and Intelligent Systems.

The rise of generative AI (e.g., ChatGPT and DALL-E) is also influencing HR. Although still emerging, generative AI tools offer new capabilities for talent management. For instance, companies experiment with AI chatbots for recruiting outreach or for coaching employees. AI can generate personalized training materials or simulate interview questions. Early evidence suggests generative AI could boost employee learning and engagement, though HR leaders must guide its responsible adoption.

Mobile and Collaboration Tech.

Another growing category is mobile HR applications and collaboration tools. Companies increasingly use mobile apps for recruiting (mobile-friendly job portals), for onboarding (apps that deliver orientation content), and for ongoing engagement (feedback apps, pulse surveys). Collaboration platforms (e.g., Teams, Slack) integrated with HR bots help in mentoring and performance check-ins. While specific studies are scarce, the general trend is that any technology improving connectivity and real-time data sharing benefits TM. Remote work technologies (video conferencing, VR training) have also expanded the talent pool and changed how skills are developed.

Talent Marketplaces and Social Tools.

Modern TM also leverages internal talent marketplaces and social networks. Internal platforms use algorithms to match employees to projects or roles across the company, fostering mobility. Social media and professional networks (LinkedIn, GitHub) serve as talent pools, with AI tools scanning these networks for potential candidates. These approaches rely on big data and AI to identify and engage talent beyond traditional channels. They reflect a shift to more dynamic, platform-based talent strategies.

In summary, emerging technologies extend the impact of TM tools. Blockchain ensures data security in recruitment; generative AI opens new avenues for content and engagement; mobile and social platforms create more agile talent ecosystems. The literature indicates that organisations experimenting with these technologies gain competitive advantages in speed, efficiency, and candidate experience. Yet, as always, implementation requires careful consideration of privacy, cost, and change management.

RECOMMENDATIONS

- Prioritise Integrated HR Platforms by investing in HRIS and digital HR platforms that centralise workforce data and integrate recruitment, retention, performance management, and learning functions.

- **Adopt Predictive Analytics for Retention:** Implement models that forecast attrition risks and skills gaps, but ensure transparency in how predictions are generated.
- **Balance AI with Human Oversight:** Use AI for efficiency in recruitment and assessment but maintain human review to safeguard fairness and trust.
- **Pilot Emerging Technologies:** Explore blockchain for credential verification and generative AI for learning and engagement, while carefully assessing scalability and costs.
- **Embed Ethical Governance:** Establish frameworks to address data privacy, fairness, and algorithmic bias, thereby increasing employee trust in technology-driven HR systems.

CONCLUSION

Technology has become inseparable from TM, shifting it from an administrative function to a strategic driver of organisational success. Research consistently shows that digital HR systems, AI, and analytics improve recruitment efficiency, employee engagement, and organisational performance (Guerra et al., 2023). Emerging tools like blockchain and generative AI promise further innovation, though their adoption must be tempered by ethical considerations.

The central challenge is not whether to use technology in TM, but which tools to prioritise. Organisations that strike the right balance—leveraging digital platforms, predictive analytics, and emerging technologies while embedding strong ethical frameworks—will be best positioned to compete globally.

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